Graph Theory & Networks (AI HL)

ANSWER KEY

Section 1: Degrees and the Handshaking Lemma

1 What does the Handshaking Lemma state for any undirected graph with V vertices and E edges?

a. **The sum of the degrees of all vertices equals twice the number of edges: $\sum \deg(v) = 2E$** ✓

b. The sum of the degrees equals E

c. The number of vertices with odd degree is always odd

d. Every vertex must have degree at least 2

Because every edge connects two endpoints, each edge contributes exactly 2 to the sum of the degrees of the vertices: $\sum \deg(v) = 2E$.

2 A graph has 6 vertices, each having degree 3. How many edges does the graph have?

a. 9 ✓

b. 18

c. 6

d. 12

$\sum \deg(v) = 6 \times 3 = 18$. By the Handshaking Lemma, $2E = 18 \Rightarrow E = 9$.

3 Why can a graph never have an odd number of vertices of odd degree?

a. **Because $\sum \deg(v) = 2E$ is always even, the sum of odd degrees must be even, forcing an even count of odd vertices** ✓

b. Because edges cannot cross each other

c. By Euler's formula $V - E + F = 2$

d. Because every vertex must have at least one adjacent edge

The total sum of degrees is even ($2E$). The sum of even degrees is even, so the sum of the odd degrees must also be even, which requires an even number of odd terms.

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Section 2: Eulerian and Hamiltonian Paths and Circuits

4 What condition is both necessary and sufficient for a connected graph to contain an Eulerian circuit (a closed trail visiting every edge exactly once)?

a. **Every vertex has an even degree ✓**

b. Exactly two vertices have odd degree

c. The graph contains no cycles

d. The graph is a complete bipartite graph

Euler proved that a connected graph has an Eulerian circuit if and only if all vertices have even degree.

5 If a connected graph has exactly two vertices of odd degree, what type of Eulerian walk does it contain?

a. **An Eulerian trail (open path visiting every edge once) that starts at one odd vertex and ends at the other ✓**

b. An Eulerian circuit returning to the start

c. No Eulerian trail or circuit can exist

d. A Hamiltonian cycle

Euler's theorem guarantees an open Eulerian trail if and only if exactly two vertices have odd degree; the trail must start at one odd vertex and terminate at the other.

6 What is the difference between an Eulerian circuit and a Hamiltonian cycle?

a. **An Eulerian circuit visits every edge once; a Hamiltonian cycle visits every vertex once ✓**

b. An Eulerian circuit visits every vertex once; a Hamiltonian cycle visits every edge once

c. They are identical concepts in graph theory

d. Hamiltonian cycles apply only to directed trees

Eulerian paths/circuits pertain to visiting every edge, while Hamiltonian paths/cycles pertain to visiting every vertex.

Section 3: Minimum Spanning Trees (Kruskal & Prim)

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7 In a connected weighted graph with V vertices, how many edges does any spanning tree contain?

a. $V - 1$ ✓

b. V

c. $V + 1$

d. $\frac{V(V-1)}{2}$

Any tree on V vertices is minimally connected and contains no cycles, having precisely $V - 1$ edges.

8 How does Kruskal's algorithm select edges to construct a Minimum Spanning Tree (MST)?

a. **Sorts all edges by weight and greedily adds the smallest available edge that does not create a cycle** ✓

b. Grows a single tree by continuously adding the closest vertex to the current tree

c. Removes the largest cycle edges one by one until only a tree remains

d. Visits vertices in alphabetical order

Kruskal's algorithm is an edge-based greedy algorithm that adds edges in order of increasing weight, rejecting any edge that completes a cycle.

9 How does Prim's algorithm differ conceptually from Kruskal's algorithm?

a. **Prim grows a single connected tree from an arbitrary starting vertex, whereas Kruskal maintains a forest of components until unified** ✓

b. Prim works only on unweighted graphs

c. Kruskal finds the longest path rather than minimum tree

d. Prim can only be applied to planar graphs

Prim's algorithm builds a single growing connected tree vertex by vertex, whereas Kruskal's algorithm connects disjoint components across the whole graph.

Section 4: Shortest Paths and Dijkstra's Algorithm

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10 What is the purpose of Dijkstra's algorithm?

a. **To find the shortest paths from a single source vertex to all other vertices in a network with non-negative edge weights ✓**

b. To find a minimum spanning tree

c. To determine if a graph has an Eulerian circuit

d. To find the maximum flow through a network

Dijkstra's algorithm is the standard greedy algorithm for computing single-source shortest paths on graphs with non-negative edge weights.

11 Why does Dijkstra's algorithm fail if a network contains negative edge weights?

a. **A greedy choice once finalized could be improved later by a negative edge, invalidating the optimality assumption ✓**

b. Distances cannot be negative by definition

c. The algorithm enters an infinite cycle immediately

d. Square roots of negative weights cannot be computed

Dijkstra's greedy assumption assumes that once a vertex distance is finalized, no shorter path to it exists; negative edges violate this monotonic property.

12 In a planar graph, if $V = 6$ and $E = 9$, how many faces F does the graph enclose according to Euler's formula $V - E + F = 2$?

a. 5 ✓

b. 3

c. 4

d. 6

By Euler's planar formula $V - E + F = 2$: $6 - 9 + F = 2 \Rightarrow -3 + F = 2 \Rightarrow F = 5$.

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